

Typhoon HIL
101

Modbus
Modbus

ECE4191

Typhoon HIL 101
real-time simulator

Integrated Design Project Laboratory Manual

Module 1

Raspberry Pi 5B

Semester 2, 2026

Battery Emulators

$$v_n = \frac{1}{2} \left(\frac{1}{n} - \frac{1}{2} \right) \cdot \left(\frac{z-1}{n} \right)^2$$

$$x_0 = P \left(1 + \frac{v_m}{z_{n-1}} \right) \longrightarrow$$

$$v_p = \frac{1}{2} (-n_k - \xi) \longrightarrow$$

$$a_n = a_0 l f + \sum_{i=1}^n (s_i + \beta(1 - a_n - a))$$

```
import python
```

```
def _init_nc(_smi:{  
    endose.conelli:1),  
    mitocetsoet);  
...)
```

```
def oostie(  
...  
def bupats()  
    return  
...)
```

```
import tbb
```

```
def qpntior(cos(ouA)):  
    sxe = holte, 8at." Pst)  
    sse(s=.nadaqun-h-s.s,T, 08.042)  
    ceturn dnr.eyeda.  
...  
def pritter4(B118:e,et):  
    return (bxrew).  
...
```

QP & MPC Algorithms

Optimisation for Smart Energy Control

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This document will be revised throughout the semester and the revision history will be summarised below.

Table 1: Revision History

Version	Date	Description of Changes
v1.0	24/07	Initial release

1 Introduction and Objectives

Welcome to the hardware implementation phase of ECE4191. This unit follows a **Interim Product Delivery (IPD)** methodology to bridge the gap between theoretical control systems and real-world power systems deployment.

The objective of Weeks 1 and 2 is to understand how to use the Typhoon HIL simulation platform and deploy a realistic distribution feeder for power system analysis. By the end of this module you will establish a fully functional **Real-Time Simulator (Typhoon HIL101)** running the IEEE 13 Node Feeder, write Python scripts that interact with the simulation through the **Typhoon HIL Python API**, and perform a five-day load and PV generation playback to analyse voltage profiles and power flows.

Learning Expectations

Module 1 aims to establish foundational knowledge on using the TyphoonHIL software along with the Monash IEEE 13 Node Feeder model.

- Setup and familiarise yourself with the Typhoon HIL software by going through the provided instructions and the tutorial from the original provider
- Implement a simple RLC circuit in the software and perform basic signal monitoring
- Understand the Monash IEEE 13 Node Feeder by exploring the nodes and sub-panels within a specific node.
- Load and analyse a 5-day real-time simulation on the Monash IEEE 13 Node Feeder.

Power vs Energy

Recall the difference between **power** and **energy**.

- **Power** is the **rate** at which energy is transferred or consumed. It describes *how fast* work is being done and is measured in:

Watts (W), Kilowatts (kW)

- **Energy** is the **total amount** of work done or stored over time. It describes *how much* energy has been used or stored and is measured in:

Watt-hours (Wh), Kilowatt-hours (kWh)

The relationship between power and energy is:

$$E = P \times t$$

where:

- E = Energy
- P = Power
- t = Time

For example, a 100 W light bulb running for 2 hours consumes:

$$E = 100 \times 2 = 200 \text{ Wh}$$

A useful analogy is:

- **Power** is like the *speed* of water flowing through a pipe.
- **Energy** is the *total amount of water* collected over time.



Figure 1: Typhoon HIL101 Real-Time Simulator. This will be used in the labs and you can access this during your lab and applied session throughout the semester.

2 Typhoon HIL

Background: What is Typhoon HIL?

Hardware-in-the-Loop (HIL) is a technique that allows engineers to test and validate real physical devices connected to a virtual representation of the wider plant (eg. power system). Controller-in-the-loop (CIL) is a technique used to test and validate real-world controllers in real time using a virtual representation of the plant. In industry, HIL and CIL are used to test, validate, and commission new equipment.

Typhoon HIL is a real-time HIL simulation platforms designed for power electronics and power systems applications. Founded in 2009, Typhoon HIL's technology centres on a proprietary **FPGA-based solver** that enables ultra-high-fidelity emulation of switching converters, electric machines, microgrids, and distribution networks [1].

The HIL101 Simulator

Elements of the **Typhoon HIL101** platform include [2]:

- **Simulation time step:** Down to 250 ns for general power electronics circuits.
- **Processing:** Up to 3 FPGA processing cores.
- **I/O:** USB 3.0, Ethernet, Gb/s serial link, JTAG, and General Purpose I/O (GPIO).
- **Built-in mixed-signal oscilloscope** for direct signal observation.

Typhoon HIL Control Center

All Typhoon HIL hardware is operated through the **Typhoon HIL Control Center**, a vertically integrated Model-Based System Engineering software toolchain. Typhoon HIL Control Center bundles four key tools [3]:

- **Schematic Editor** — A graphical environment for building power electronics and power system models from a rich component library. Once a schematic is complete, it can be compiled and deployed to the HIL hardware.
- **HIL SCADA** — A real-time supervisory control and data acquisition interface. You can drag and drop customisable widgets (scopes, sliders, buttons) to visualise signals and interact with the running simulation.
- **TyphoonTest IDE** — A Python-based test automation framework for writing and executing repeatable test scripts.
- **Virtual HIL** — A software-only emulator included with every Typhoon HIL Control Center installation, allowing model development and offline simulation without physical hardware.

Applications and Industry Relevance

Typhoon HIL platforms are used across a broad range of domains relevant to modern electrical engineering [4]:

- **Power electronics:** Validation of inverter, DC/DC converter, and motor drive controllers.
- **Grid modernisation:** Testing of microgrid controllers, BESS, and digital substations.
- **Electric vehicles (e-Mobility):** Battery Management Systems (BMS), on-board chargers, and EV charging station controllers.

- **Academic research:** Rapid prototyping and testing of novel control algorithms to minimize the risk and cost of physical hardware.

In this module, you will use the HIL101 to simulate a realistic distribution network (the IEEE 13 Node Feeder) and interact with it entirely from the Windows host using the **Typhoon HIL Python API** — reading simulated grid measurements and writing control setpoints without any external hardware.

3 Typhoon HIL Setup

Operating System Requirements

The **Typhoon HIL Control Center** software provided in Moodle is intended for **Windows** machines with **Intel/AMD CPU**. Linux and Mac users are recommended to use the **Monash Online Virtual Environment (MoVE)** platform for Typhoon HIL Control Center operation. This can also interface with the physical devices, but we still recommend installing the application on a Windows device for ease of use.

Step 1: Software Installation using your personal laptop

1. **Sign Up:** Create an account on the [Typhoon HIL Subscription Platform](#) using your **Monash student email** (xxxxxxx@student.monash.edu) and the activation key provided in Moodle. This account is required for licensing and software access.
2. **Download:** Download the **Typhoon HIL Control Center** software from the **Moodle page**.
3. **Install:** Run the installer.
 - *Note:* Installation takes approximately 10 minutes to complete. A full installation requires approximately 8GB of disk space but we recommend having more storage for additional files that you will need later on.

Step 2: Licensing and Activation

The HIL101 requires a specific hardware license to run real-time simulations.

1. Open **Typhoon HIL Control Center** from your desktop.
2. Once prompted (this will appear only once after a fresh installation), select **Do not import settings** and click **OK**, as shown in Figure 2.
3. Log in via the user icon in the top-right corner as shown in Figure 3, using your Monash credentials to pull your basic Virtual HIL license .
4. Navigate to **License** → **Download License** as shown in Figure 4. Once the import is successful, restart HIL Control Center to finalize the activation and apply all changes.
5. Enter the Unit Activation Key below (with hyphens and no spaces) that will be given in Moodle.
6. Upon successful login and license activation, both the  **License** and  indicators will appear in green.

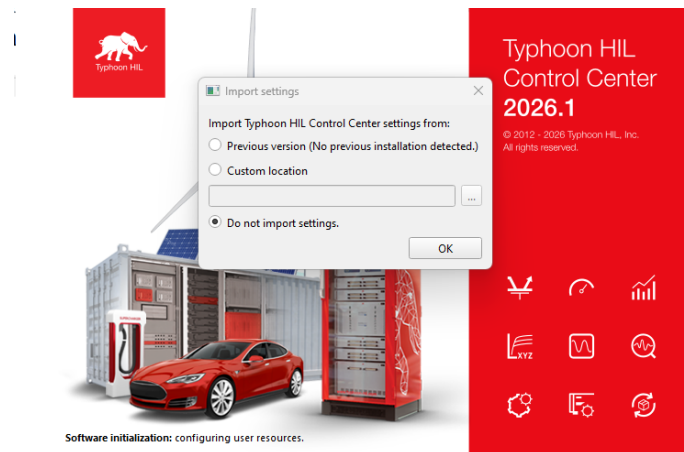


Figure 2: Initial setup prompt in Typhoon HIL Control Center to import settings

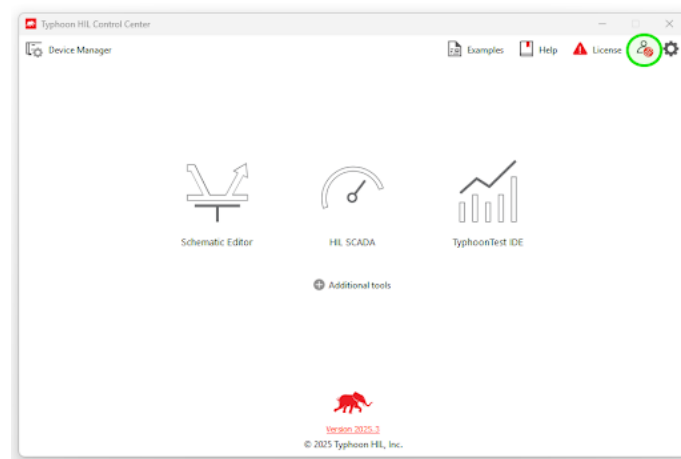


Figure 3: Typhoon HIL Control Center - Welcome Screen

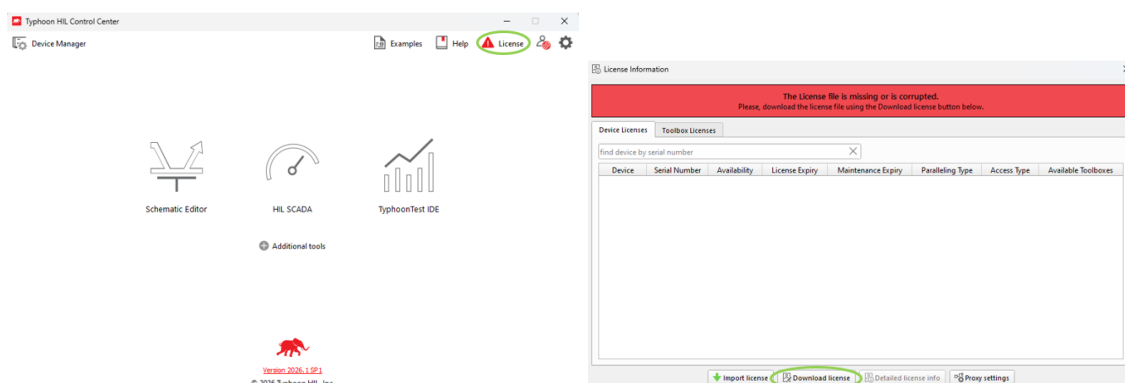


Figure 4: Update License

Step 3: Firmware Synchronisation

If the Typhoon HIL software on your laptop is older than the firmware on the HIL101, you must sync them.

1. Ensure the DC power supply is connected to the HIL101 using the DC adaptor and ensure it has been powered on at the wall outlet.
2. Connect the HIL101 to your computer using the provided high-speed USB-B to USB-A cable. For those who do not have a standard USB-A jack on their laptops, you will need a USB adaptor.

- Start the HIL101 unit by clicking the red  power button. Upon booting up, the **White** LED on the front panel of the physical HIL101 will blink a few times and settle.
- Click on **Device Manager** in the main Typhoon HIL Control Center window as shown in Figure 5.

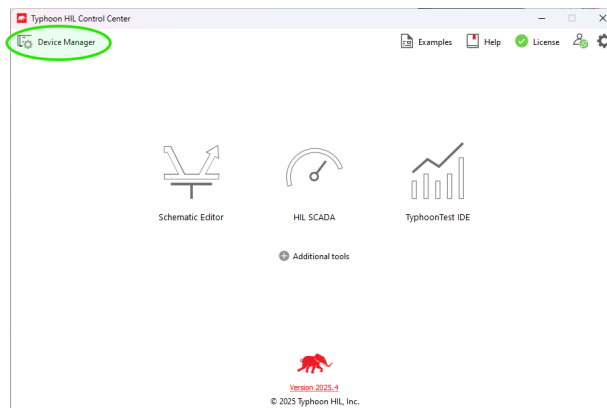


Figure 5: Device Manager

- If Device Manager does not detect any HIL device, click **Refresh**. The HIL101 unit should then appear in the **Detected devices** sub-panel, as shown in Figure 6.

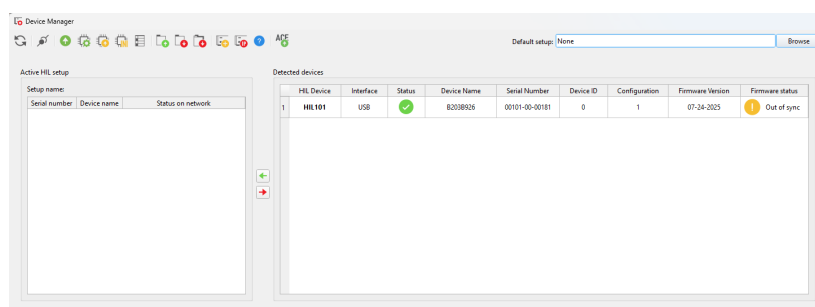


Figure 6: HIL101 detected

- Ensure that **HIL Device** is set to **HIL101** and **Configuration** number set to **1**. Other settings can be ignored.
 - Right-click the device and select **Sync firmware**.
- WARNING:** Do not disconnect the USB cable during the update process.
- Once the firmware has been synchronised, the unit will reboot automatically.
 - The Firmware status should now show as **Synced**.

Exercise

Objective: You must demonstrate the simulation of a basic RLC circuit in real-time.

Task Description: Create a Series RLC circuit.

Action Items:

1. Open **Schematic Editor**  from the Typhoon HIL Control Center Welcome Screen.
2. To create a new schematic, go to **File** → **New** → **New model ...**
3. Set the new schematic settings as shown in Figure 7. Ensure that **Hardware product** is set to **HIL101** and **Configuration** set to **1**.

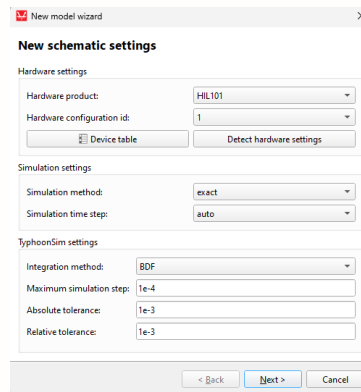


Figure 7: New schematic settings

4. Drag and drop a Voltage Source, Resistor, Inductor, and Capacitor from the library.
5. Build the circuit as shown in Figure 8. Connect the Current Measurement block in series with R_{in} , and place the Voltage Measurement block in parallel with R and C .

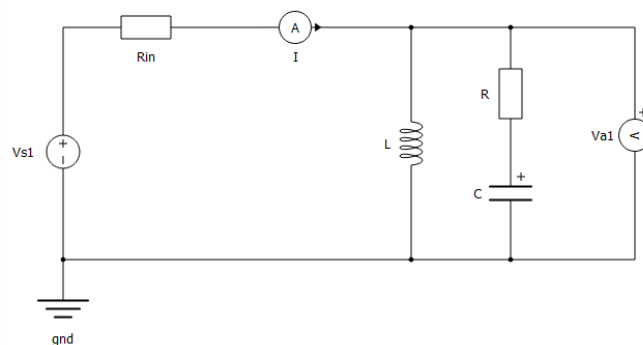


Figure 8: RLC Circuit

6. Set V_{s1} as: Source nature = Sine, RMS voltage = 350 V, Frequency = 3000 Hz, Phase = 0° . Set $R_{in} = 5 \Omega$, $L = 1 \times 10^{-5} \text{ H}$, $R = 0.1 \Omega$, $C = 5 \times 10^{-4} \text{ F}$.
7. Save the schematic.
8. In the Schematic Editor, click  **Compile and (re)load model in HIL SCADA**.
9. Once compilation is complete, return to the **Typhoon HIL Control Center Welcome Window** and open  **HIL SCADA**.

10. Click **Create new Panel**.

11. In the SCADA window:

- Drag a **Capture/Scope** widget from the Library Explorer onto the panel as shown in Figure 9.

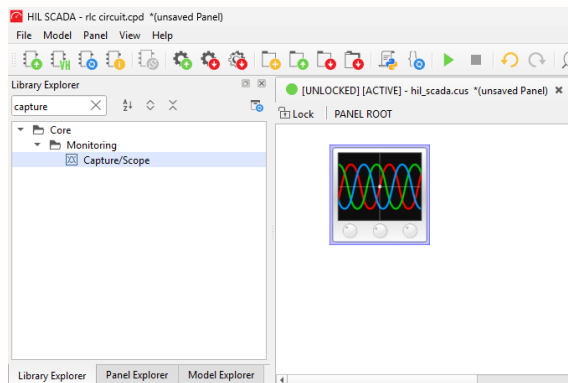


Figure 9: Adding Capture/Scope to panel

- Double click on the **Capture/Scope**  widget to open **Capture/Scope Windowed Mode**.
- Click on **Signals** in **Capture/Scope Windowed Mode** as shown in Figure 10.

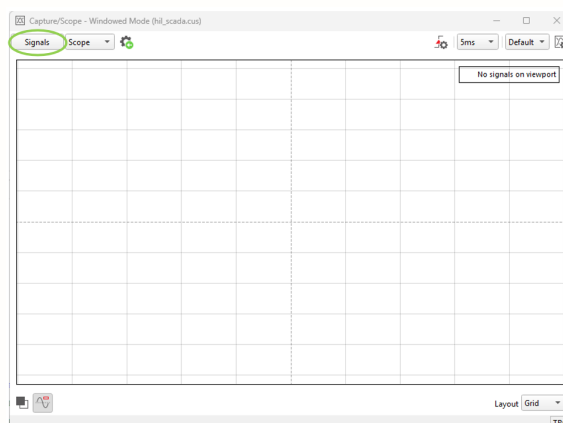


Figure 10: Capture/Scope Window

- Click on the  **Plus** symbol to assign V_{a1} and I signal to the Scope as shown in Figure 11.

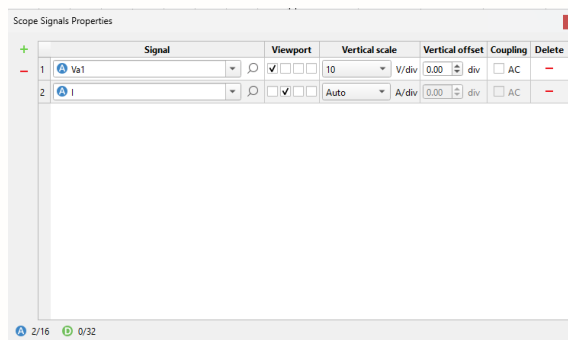


Figure 11: Setting of scope with input and output signals

- Click the  **Simulation** button in the HIL SCADA toolbar to start the real-time simulation.
- Set the plot time step to $200\mu s$. This can be changed from the top-right corner of the Scope window, as shown in Figure 12.
- The waveform on the scope will appear as shown in Figure 12.

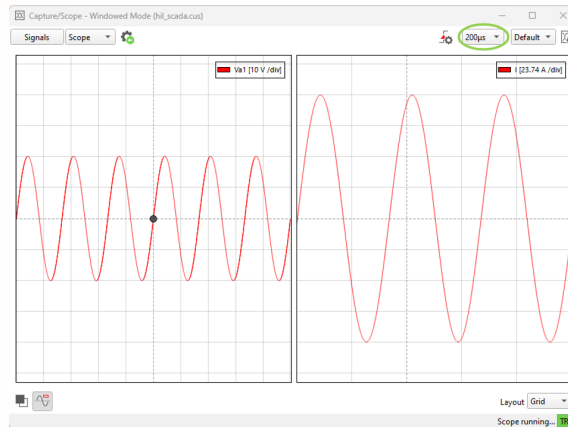


Figure 12: Observed waveform on the Scope

^aThere might be a deprecation warning here which is fine to proceed.

Further Learning: Explore Typhoon HIL Control Center Examples and HIL Academy

Typhoon HIL includes a built-in library of example models covering a wide range of applications — from simple communication setups to battery inverter control and microgrid management. These examples are an excellent starting point for understanding how real models are structured in the Schematic Editor and HIL SCADA.

To access the examples: From the Typhoon HIL Control Center Welcome Window, click **Examples** in the top menu bar as shown in Figure 13. Browse the categories or use the search bar to find models relevant to your interests.

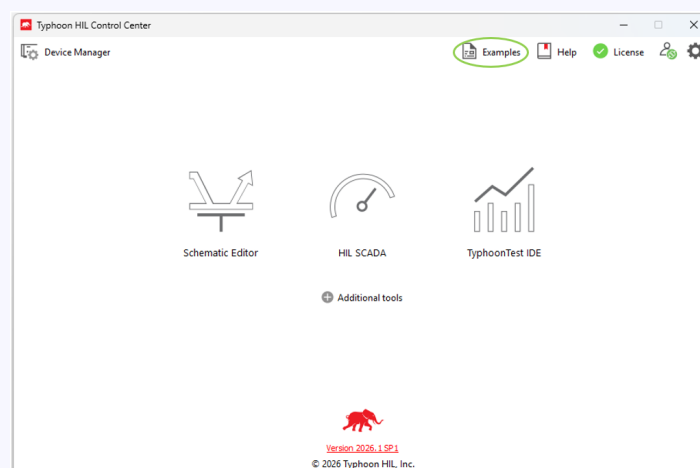


Figure 13: Accessing the built-in example models from the Typhoon HIL Control Center Welcome Window via the **Examples** button (circled).

Watch: HIL Fundamentals Course

For a structured introduction to HIL simulation concepts and the Typhoon HIL platform, students are

strongly encouraged to complete the free **HIL Fundamentals** course available on the HIL Academy:

<https://hil.academy/courses/hil-fundamentals/>

The course covers real-time simulation principles, the Typhoon HIL Control Center toolchain, schematic modelling, HIL SCADA operation, and testing workflows — all directly relevant to the tasks in this module.

4 IEEE 13 Node Feeder Setup

What Are IEEE Test Feeders?

The **IEEE Distribution Test Feeders** are a set of standardised benchmark networks developed and maintained by the IEEE Power Engineering Society (PES) Distribution System Analysis Subcommittee. Originally introduced in 1992, these test feeders were designed to provide researchers and software developers with reduced-order models of real distribution circuits, enabling consistent evaluation and benchmarking of power flow algorithms, control strategies, and optimisation methods for unbalanced three-phase radial distribution systems [5].

The **IEEE 13 Node Test Feeder** is one of the most widely used feeders in the IEEE PES test feeder suite. It operates at a nominal distribution voltage of **4.16 kV**, and includes both single phase and two-phase lateral [6]. The single-line diagram of the feeder is shown in Figure 14.

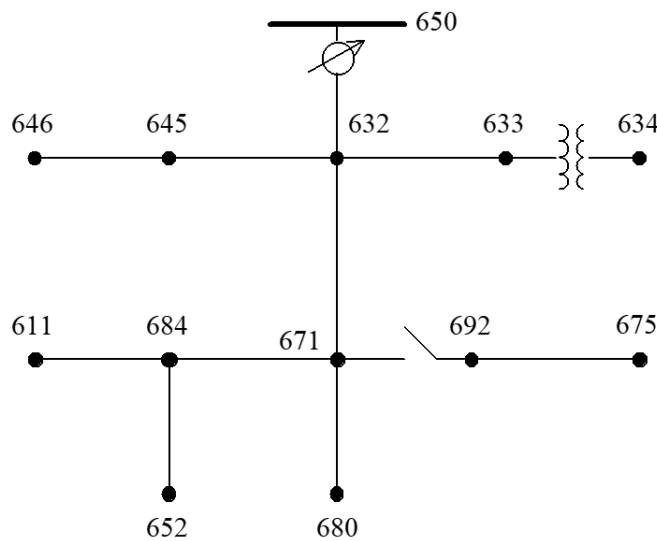


Figure 14: Single-line diagram of the IEEE 13 Node Test Feeder [7]. The feeder consists of 13 nodes interconnected by overhead and underground line segments, with a substation voltage regulator, two shunt capacitor banks, an in-line step-down transformer, and a mix of spot and distributed loads across single-, two- and three-phase laterals.

The key characteristics of the feeder are summarised below [5, 7]:

- **Network size:** 13 nodes connected by a combination of overhead and underground line segments with varying one-, two-, and three-phase configurations.
- **Loading:** Short and relatively heavily loaded for a 4.16 kV feeder, with both unbalanced time-varying loads and distributed loads across the network.
- **Voltage regulator:** One substation voltage regulator connected to node 650, used to regulate voltage profiles along the feeder.
- **Shunt capacitor banks:** Two Capacitor Banks (CBs) providing Reactive Power (VAr) compensation — a three-phase bank at node 675 and a single-phase bank at node 611.
- **Supply Transformer:** A 4.16 kV/0.48 kV step-down transformer between node 633 and 634.
- **Unbalanced conditions:** Unbalanced three-phase loading, single-, two- and three-phase laterals, a realistic representation of an unbalanced Low Voltage (LV) and Medium Voltage (MV) distribution feeder.

Typhoon HIL IEEE 13 Node Test Feeder

Typhoon HIL includes its own pre-built implementation of the IEEE 13 Node Test Feeder in the Typhoon HIL Control Center Examples library. It can be found by searching **13 node** in the Examples browser, as shown in Figure 15.

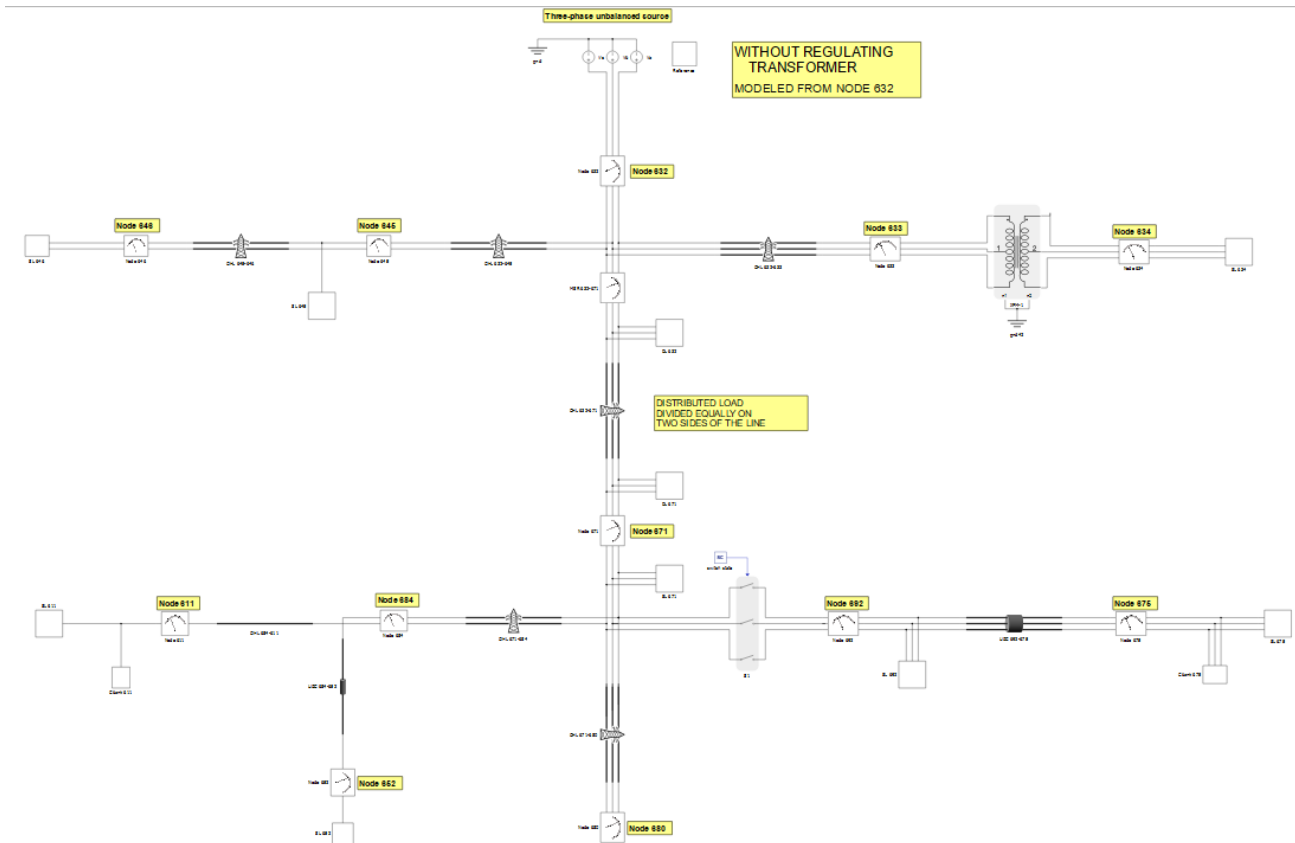


Figure 15: Typhoon HIL's built-in IEEE 13 Node Test Feeder example, accessible via **Examples** in the Typhoon HIL Control Center Welcome Window.

Important: Do Not Use the Typhoon Example Feeder for ECE4191

The Typhoon HIL built-in IEEE 13 Node Test Feeder, as shown in Figure 15, is a **standard implementation** of the IEEE 13 Node benchmark. This implementation does not include time-varying loads or a Python API communication interface as is required in this unit. **In ECE-4191, you must use the Monash IEEE 13 Node Feeder (sometimes abbreviated as MIEEE-13NF) as provided on Moodle**, which is described in the following sections.

The Monash IEEE 13 Node Feeder (MIEEE-13NF)

In this unit, the IEEE 13 Node Feeder has been modified and developed for ECE4191 — we call the modified version the **Monash IEEE 13 Node Feeder**. The MIEEE-13NF has been implemented in the Typhoon HIL Schematic Editor (`Monash_13NodeFeeder.tse`) and includes custom time-varying load components packaged as a user library (`Monash_library_time_varying_load.tlib`). The compiled model (`Monash_13NodeFeeder.cpd`) is loaded directly into HIL SCADA, where a panel (`Monash_13NodeFeeder.cus`) provides real-time visibility of node voltages, line currents, and power flows across the feeder.

In control engineering terminology, the MIEEE-13NF is the **plant** and in ECE4191 you will develop **controllers** for this plant. In Module 1, you will identify operational problems in the plant that motivate the need to develop controllers (in Module 2, you will be introduced to Model-Predictive Controllers). More

specifically, in Module 1, Python scripts running on the Windows host interact with the simulation using the **Typhoon HIL Python API** — reading measurements from feeder nodes and writing signals back in real time. In this way, you will investigate instances of over-voltage and under-voltage events across the MIEEE-13NF over a 5-day period.

Comparison between the IEEE 13 Node Feeder and Monash IEEE 13 Node Feeder

Key Features of the MIEEE-13NF

The table below summarises the key differences between the standard IEEE 13 Node Feeder and the Monash IEEE 13 Node Feeder.

Table 2: Comparison between the standard IEEE 13 Node Feeder and the Monash custom implementation.

Feature	Standard IEEE 13 Node	Monash Version
Source voltage (line-to-ground)	$V_a = 2452.2 \text{ V}$, $V_b = 2502.65 \text{ V}$, $V_c = 2443.57 \text{ V}$	$V_a = 2461 \text{ V}$, $V_b = 2461 \text{ V}$, $V_c = 2461 \text{ V}$
Load model	Fixed constant impedance (Spot Load)	Time-varying controllable via SCADA/Python API
PV generation	None	At every time-varying load node
Node 671 implementation	Standard single-CPU load	Dual-CPU partitioned for stability at high load
Node 634 implementation	N/A	Master Pulse mechanism at Node 634
External communication	None	Accessible via Typhoon HIL Python API
SCADA interface	Generic scope only	Custom per-node sub-panels with voltage/power trace plots
Playback launcher	N/A	Auto-launches on simulation start

5 Monash IEEE 13 Node Feeder Setup

Step 1: Download and Extract Files

1. Download the file **Monash IEEE Feeder Model.zip** from the **Moodle page**.
2. Extract (unzip) the contents into a working directory on your computer.

Step 2: Open Typhoon HIL Schematic Editor

1. Open **Typhoon HIL Control Center** from the Start Menu.
2. Click on **Schematic Editor**.

Step 3: Set Library Path

1. In the Schematic Editor, navigate to:

File → Modify Library Paths...

2. Click **Browse** and select the folder where you extracted the files.
3. Ensure the following options are enabled:
 - **Load**
 - **Include Subdirectories**
4. Click **Apply** and then **Save**.
5. This process is illustrated in Figure 16.

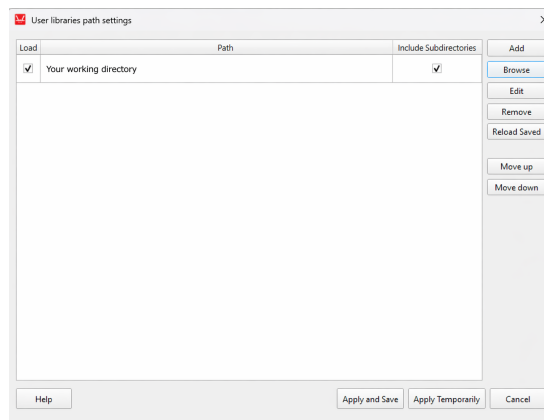


Figure 16: Setting library path in Typhoon HIL Schematic Editor

Step 4: Reload Libraries

1. Navigate to:

File → Reload Libraries

Step 5: Open the Feeder Model

1. Navigate to:

File → Open... → Open Models...

2. In the file explorer window, navigate to your **Working Directory** containing the extracted feeder model files.
3. Select the file:

Monash_13NodeFeeder.tse

4. Click **Open**.
5. Once the model is successfully loaded, the IEEE 13 Node Feeder schematic should appear in the Schematic Editor, as shown in Figure 17.

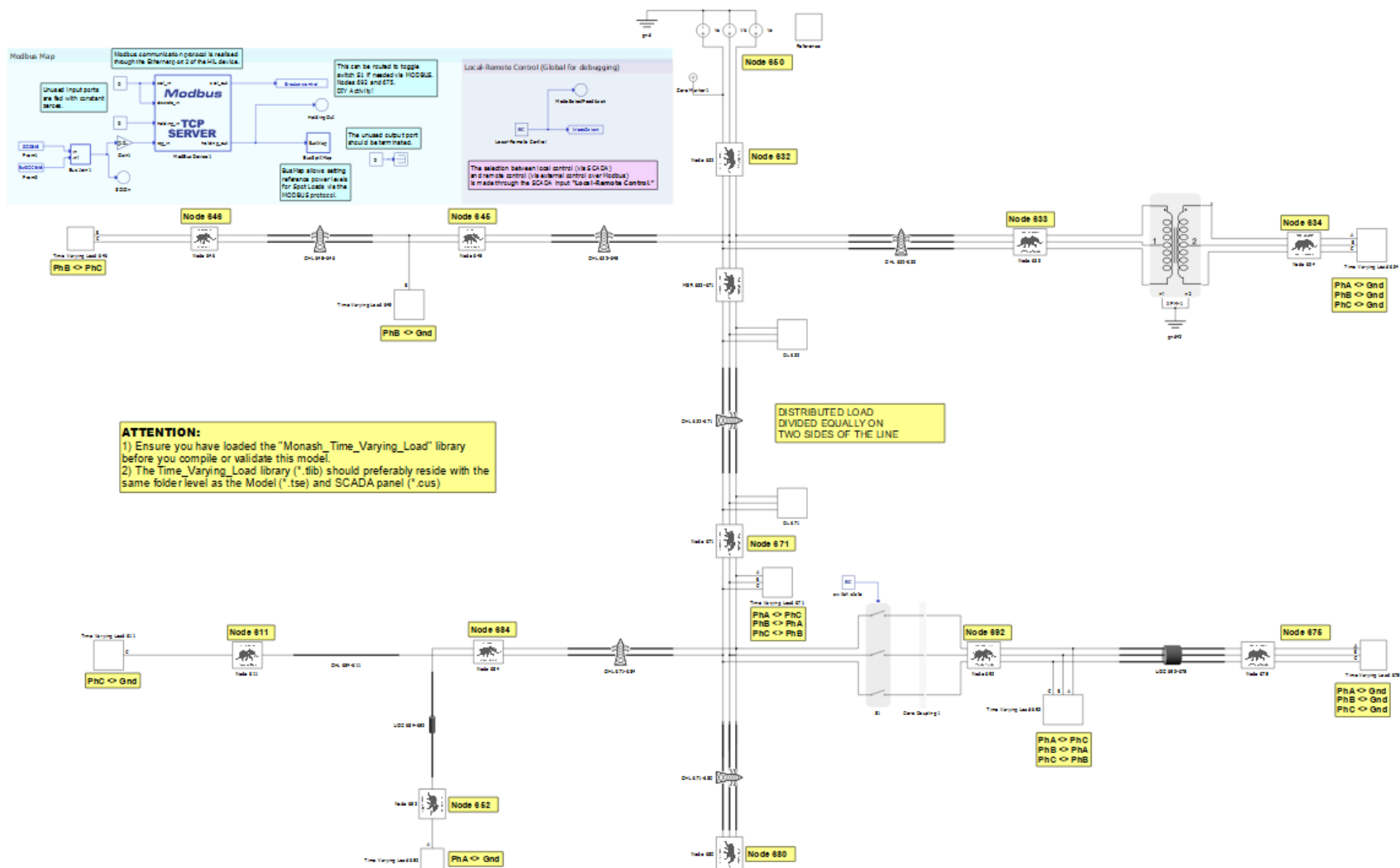


Figure 17: Monash IEEE 13 Node Feeder schematic loaded in the Typhoon HIL Schematic Editor.

The schematic is organised into the following regions:

- **Python API Map** (top-left, blue region) — contains the Modbus TCP Server block and the BusSplitMap subsystem. All feeder measurement signals are routed here. In Module 1 these signals are accessed via the Typhoon HIL Python API running on the Windows host. A **Local–Remote Control** toggle switches between manual SCADA entry (Local Control) and programmatic writes (Remote Control).
- **Three-phase unbalanced source** (top-centre) — the 650 node supplies the feeder, consisting of three single-phase sinusoidal voltage sources (V_a , V_b , V_c) operating at 60 Hz, connected to a **Reference** ground subsystem. We can change the voltages of the voltage sources to represent the control action of a voltage regulator corresponding to the substation transformer.
- **Time-Varying Load nodes** (634, 645, 646, 652, 611, 671, 675, 692) — each node contains one or more `Monash_time_varying_load` custom component subsystems. Each time-varying load accepts real-time `Pref`, `Qref`, `Battery_Power`, and `PVpanel_power` SCADA inputs.

Attention: Node 634 — Master Pulse Required

The Monash IEEE 13 Node Feeder includes a prominent annotation: *“THIS IS THE ONLY NODE THAT REQUIRES THE MASTER PULSE TIME-VARYING LOAD MODEL TO SUPPORT REAL-TIME SIMULATION!”* Node 634 uses the **Single phase** time-varying load with the **Master Pulse** variant for all three phases.

- **Distributed loads** — the yellow annotation *“DISTRIBUTED LOAD DIVIDED EQUALLY ON TWO SIDES OF THE LINE”* marks the DL 671 and DL 632 fixed resistive load subsystems. These represent the uniformly distributed load along each line section, split equally between the sending and receiving ends of the segment.
- **In-line step-down transformer** (Node 633–634) — reduces the feeder nominal voltage from 4.16 kV to 480 V to supply the load at the Node 634 load section, visible as the transformer symbol on the 633–634 branch.

Step 6: Compile and Run the Model in HIL SCADA

1. Ensure the HIL device is powered on and connected to your PC/laptop via USB.
2. In the Schematic Editor, click  **Compile and (re)load model in HIL SCADA**.
3. Once compilation is complete, return to the Typhoon HIL Control Center Welcome Window and open **HIL SCADA**.
4. In the HIL SCADA window, navigate to:

File → Load Model

5. Navigate to your Working Directory, then open the folder:

Monash_13NodeFeeder Target files

6. Select the compiled model file:

Monash_13NodeFeeder.cpd

Click **Open** to load the compiled model.

7. After loading, locate the *Panels files found in the Model Directory* section in the HIL SCADA window and double-click:

`Monash_13NodeFeeder.cus`

The feeder interface will open as shown in [Figure 18](#).

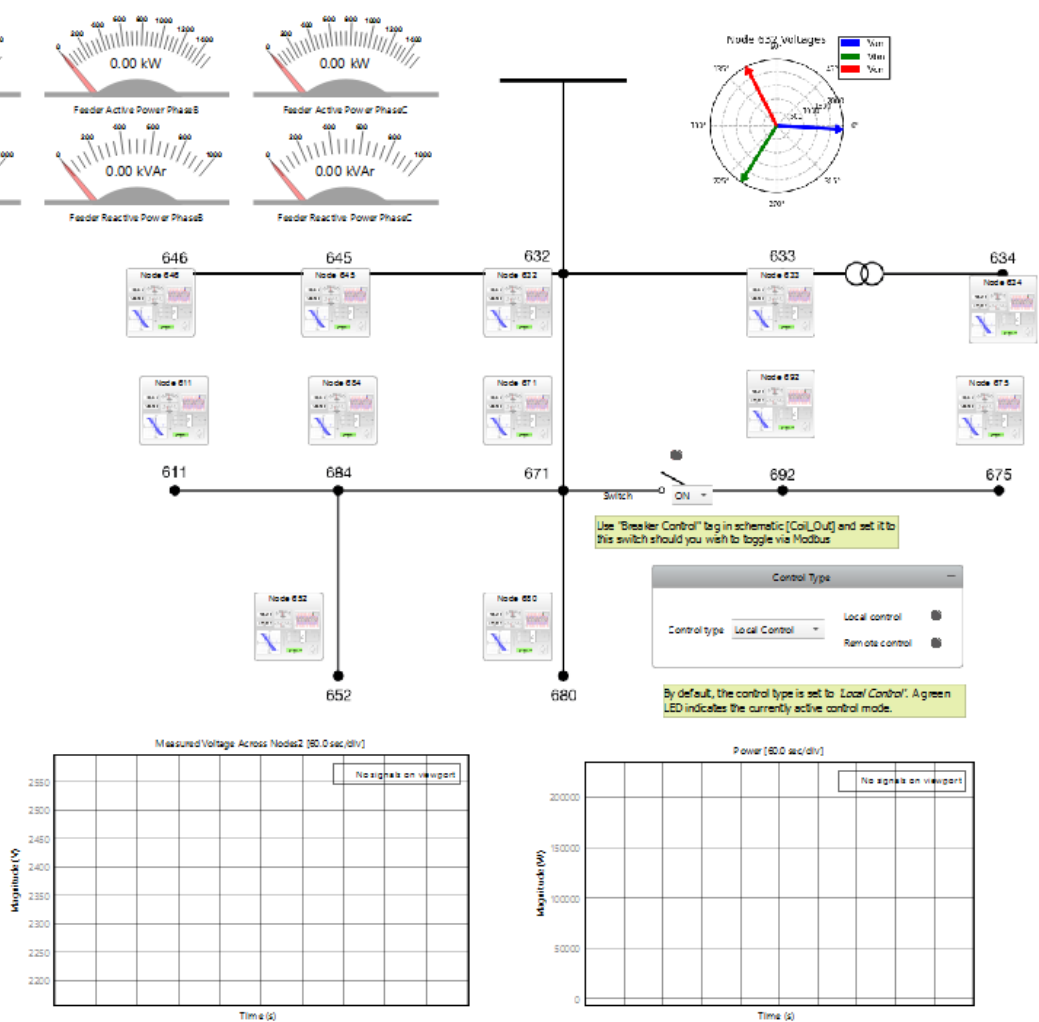


Figure 18: Monash IEEE 13 Node Feeder represented in HIL SCADA as a feeder topology diagram, showing the topology, per-node sub-panels, feeder power gauges, and control type selector.

The main SCADA panel (Figure 18) displays the following elements:

- **Feeder topology diagram** (centre) — a single-line representation of the 13-node network with all active nodes labelled (632, 633, 634, 645, 646, 671, 675, 692, 652, 611, 684, 680). Each node is shown at its correct electrical position in the radial feeder.
- **Per-node sub-panels** — a small instrument panel is embedded at each node, displaying live three-phase voltage and current readings. Double-clicking any node sub-panel opens the full detailed view for that node.
- **Feeder power gauges** (top-left) — six analogue gauge instruments display the three-phase Active Power (kW) and Reactive Power (kVAr) measured at the feeder head (Node 632), updated in real time during simulation.
- **Phasor diagram** (top-right) — a polar plot showing the voltage phasors for all three phases at the feeder reference bus, providing a visual indication of phase balance and angle separation.
- **Control Type selector** (right panel) — a toggle between **Local Control** and **Remote Control** mode. In Local Control mode, setpoints are entered manually via the SCADA panel. In Remote Control mode, setpoints are received from the Typhoon HIL Python API running on the Windows host. A green LED indicator shows which mode is currently active.
- **Feeder breaker switch** — located between node 671–692, this switch can be toggled to open or close during a real-time simulation.
- **Measured Voltage Across Nodes plot** (bottom-left) — a real-time TraceDisplay widget plotting RMS voltage (V) versus time for seven node-phase signals: Node 645.V2_rms, Node 646.V2_rms, Node 671.V1_rms, Node 671.V2_rms, Node 671.V3_rms, Node 692.V3_rms, and Node 611.V3_rms.
- **Power plot** (bottom-right) — a real-time TraceDisplay widget plotting measured active power (W) versus time for nine node-phase load signals spanning Nodes 645, 646, 652, 671 (all three phases), and 675 (all three phases).

8. Double-click on **Node 646** in the feeder topology diagram. The Node 646 detailed sub-panel will open, as shown in Figure 19.

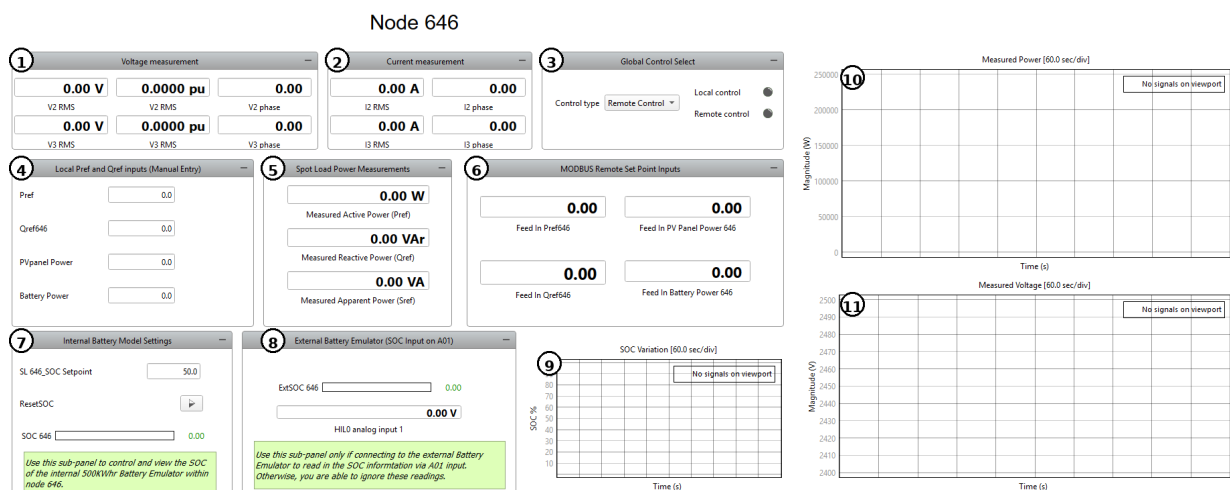


Figure 19: Node 646 detailed sub-panel in HIL SCADA, showing voltage and current measurements, local and remote setpoint inputs, BESS State of Charge (SoC) tracking, and the measured power plot.

The Node 646 sub-panel (Figure 19) is the most instrumented node in the feeder. It consists of the following functionality displays:

- **Label 1: Voltage measurement** — displays V_2 and V_3 RMS voltages in volts and per-unit, together with their phase angles.
 - **Label 2: Current measurement** — displays I_2 and I_3 RMS currents and their phase angles.
 - **Label 3: Global Control Select** — switches Node 646 between **Local Control** and **Remote Control**.
 - **Label 4: Local Pref and Qref inputs** — provides manual inputs for Pref, Qref646, PV power, and battery power.
 - **Label 5: Spot Load Power Measurements** — displays the measured active, reactive, and apparent power at Node 646.
 - **Label 6: MODBUS Remote Set Point Inputs** — displays the active-power, reactive-power, PV, and battery setpoints received through MODBUS.
 - **Label 7: Internal Battery Model Settings** — provides the SoC setpoint, SoC reset control, and current internal battery SoC.
 - **Label 8: External Battery Emulator** — displays the external SoC received through HIL analogue input A01. This can be ignored when the internal battery model is used.
 - **Label 9: SoC Variation plot** — displays the internal battery SoC over a rolling 600-second window.
 - **Label 10: Measured Power plot** — displays the measured active power at Node 646 over a rolling 600-second window.
 - **Label 11: Measured Voltage plot** — displays the phase-2 RMS voltage, V_2 , over a rolling 600-second window.
9. Click the  **Simulation** button in the HIL SCADA toolbar to start the real-time simulation. Once running, the status indicator in the top-right corner of the panel will change to **ACTIVE** (shown in green). Observe the voltage and current readings update across all node sub-panels, and watch the Measured Power Plot and SoC Variation Plot at Node 646 begin to trace live data.
10. When finished, click the  **Stop** button to halt the simulation. The **ACTIVE** indicator will return to its idle state.

6 Assessment (10% of your final grade for ECE4191)

As part of your assessment for Module 1, show your **real-time simulation** results for the voltages and power flows across the Monash IEEE 13 Node Feeder and provide oral responses. Specifically, answer questions based on your voltage and power flow simulation results. You have 10 minutes to complete this assessment.

It is recommended that you time yourself ahead of completing this assessment to ensure you have sufficient time to complete the real-time simulation demonstration and answer the questions below. We also recommend recording your real-time simulation results ahead of time so that you can study the results and prepare to answer the following questions.

Real-time Simulation Results (50% of your assessment)

You will present three real-time simulation results for the 5-day playback.

1. Real-time RMS line-to-ground voltages for each single-phase node operating at the nominal 2401 V .
2. Real-time RMS line-to-ground voltages for each single-phase node operating at the nominal 277 V.
3. Real-time active power at all single-phase nodes with time-varying load.

Oral-based questions (50% of your assessment)

Your answers to these questions should be verified using the real-time simulation plots, corresponding to the 5-day playback. Your answers need to be completed during your 10-minute allotted time.

1. Which nodes violate the upper or lower voltage bounds?
2. When do the voltage violations occur?
3. What is the worst-case voltage violation for each node where a voltage violation occurs?
4. Which nodes exhibit reverse active power flow?
5. When does the reverse power flow occur?
6. What is the largest reverse power flow for each node where reverse power flow occurs?
7. When is the total feeder net load highest across the five-day playback?
8. Which nodes contribute most to the peak load for the feeder?
9. How does the peak load period influence voltages across the feeder?

Assessment: $P_{\text{load}} + P_{\text{PV}}$ Time-Series Playback via the Typhoon API

Objective: Using the Monash IEEE 13 Node Feeder, import time-varying load and PV generation data for each of the 16 controllable nodes, considering a five-day period starting in 7 January 2013. Observe voltages at each node and power flows across the feeder. Identify feeder conditions that create over-voltage and under-voltage events. In ECE 4191, measured voltages greater than, or less than, 5% of the nominal voltage are termed ‘over-voltage’ and ‘under-voltage’ events, respectively.

In this assessment, you will run your simulation in real-time. We encourage you to prepare for the assessment by running your simulation ahead of time and identifying important features (e.g., voltage violations and the root cause of the voltage violation).

Background: $P_{\text{load}} + P_{\text{PV}}$ Dataset: 7-12 January 2013

The file `agg_jan2013_students.csv` contains aggregated half-hourly load and PV generation profiles for five consecutive summer days in January 2013. This data is to be used for the assessment tasks in Modules 1-3. Each row represents one 30-minute interval for a single phase node. Note that the disaggregated load and PV generation data was made available by Ausgrid, where smart meters recorded residential load and PV generation data separately.

Dataset structure (160 rows $\times$ 53 columns):

Column	Description
Date	Calendar date (7-Jan-13 to 11-Jan-13)
Node	Monash IEEE 13 Node feeder: Nodes with time-varying load + PV (646, 645, 611, 652, 671, 692, 675, 634)
Phase	Phase label (Ph1, Ph2, Ph3, A, B, or C)
N_Customers	Number of customers aggregated at this node-phase
Profile	Either GC_Load_kW or PV_Generation_kW
0:30 ... 0:00	48 half-hour columns (in kW, average power per 30-min interval)

Net Load (P_{net}): The net load at each node on the Monash IEEE 13 Node Feeder is:

$$P_{\text{net}} = P_{\text{load}} - P_{\text{PV}},$$

where P_{load} represents the total load consumed by customers connected to the node, and P_{PV} represents the total PV generation produced by customers connected to the node.

Note on Node 692. There are no customers connected to phases B and C, accordingly, there is no load drawn by customers on these phases.

Files required (download from Moodle):

- `p_v_load_windows_typhoon_students.py` — the playback script
- `agg_jan2013_students.csv` — the five-day load + PV dataset

Place both files in the **same folder** as `Monash_13NodeFeeder.tse`, `Monash_13NodeFeeder.cus`, and the `.tlib` library. This folder is referred to as your **Project Directory**.

Prerequisites

1. The Monash IEEE 13 Node Feeder model (`Monash_13NodeFeeder.tse`) is compiled and loaded in HIL SCADA.
2. HIL SCADA is open and the panel `Monash_13NodeFeeder.cus` is loaded.
3. The HIL101 is powered on and connected via USB.
4. **Do not start the simulation yet.**

1. Update the path variables in the panel initialization code.

The `Monash_13NodeFeeder.cus` panel already contains the playback launcher in its **initialization code**. You do **not** need to add any new code — you only need to update two path variables to match your machine.

In HIL SCADA, click on  **Open Panel initialization dialog**. Locate the following block near the top of the existing initialization code:

```

1 import os
2 import subprocess
3
4 # ----- UPDATE THESE TWO PATHS FOR YOUR MACHINE -----
5 # SETTINGS_DIR: full path to the folder containing the
6 #               playback script and CSV dataset
7 SETTINGS_DIR = r"C:\path\to\your\Project Directory"
8
9 # TYPHOON_PYTHON: full path to the Typhoon HIL Python
10 #                executable (check your Typhoon HIL Control Center version)
11 TYPHOON_PYTHON = r"C:\Program Files\Typhoon HIL Control Center 2026.1 sp1\bin\
12                  typhoon-python.cmd"
13 # -----

```

Listing 1: Path variables to update in the existing Panel Initialization code.

Replace both placeholder values with the correct paths for your setup.

Important: Correct Paths

- **SETTINGS_DIR** — the full Windows path to the folder containing `pv_load_windows_typhoon_students.py` and `agg_jan2013_students.csv`. Use a raw string (`r"..."`) to handle backslashes correctly.
- **TYPHOON_PYTHON** — the path to `typhoon-python.cmd` in your Typhoon HIL Control Center installation. The version number in the path (e.g., `2026.1 sp1`) must match your installed version. To find it, open File Explorer and navigate to:
`C:\Program Files\Typhoon HIL Control Center XXXX.X\bin\`

2. Save and close the Panel Initialization dialog.

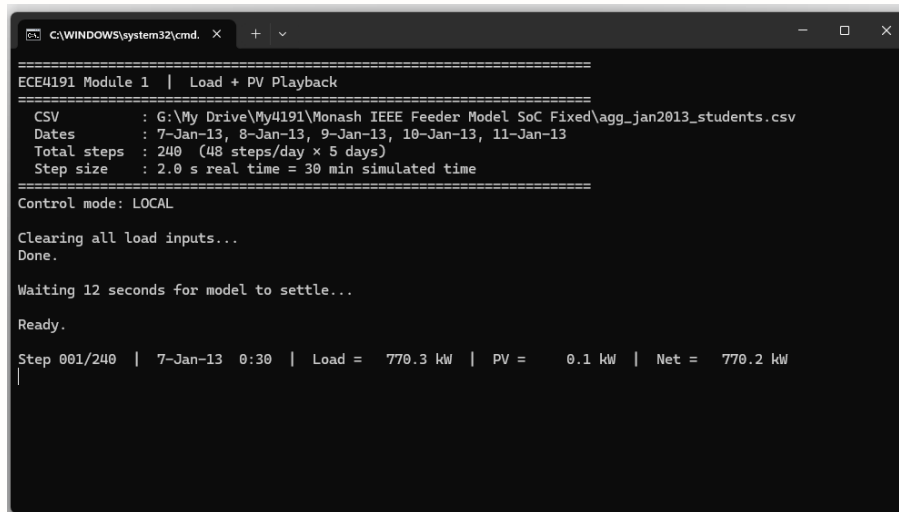
Click **OK** to save and close. The rest of the initialization code (the `subprocess.Popen` call) is already present and will launch `pv_load_windows_typhoon_students.py` automatically when the simulation starts.

3. Ensure Local Control mode is selected.

In the SCADA panel, locate the **Control Type** selector and confirm it is set to **Local Control**. The **Local control** green LED should be illuminated. The playback script writes directly to SCADA Input signals, which requires Local Control mode to be active.

4. Start the simulation.

Click the  **Simulation** button in the HIL SCADA toolbar. The Panel Initialization code runs automatically at simulation start, launching `pv_load_windows_typhoon_students.py` as a background process. A console window will appear on your Windows desktop showing playback progress, as shown in Figure 20.



```

C:\WINDOWS\system32\cmd. X
=====
ECE4191 Module 1 | Load + PV Playback
=====
CSV       : G:\My Drive\My4191\Monash IEEE Feeder Model SoC Fixed\agg_jan2013_students.csv
Dates      : 7-Jan-13, 8-Jan-13, 9-Jan-13, 10-Jan-13, 11-Jan-13
Total steps : 240 (48 steps/day x 5 days)
Step size   : 2.0 s real time = 30 min simulated time
=====
Control mode: LOCAL

Clearing all load inputs...
Done.

Waiting 12 seconds for model to settle...

Ready.

Step 001/240 | 7-Jan-13 0:30 | Load = 770.3 kW | PV = 0.1 kW | Net = 770.2 kW

```

Figure 20: Console output from `pv_load_windows_typhoon_students.py` after simulation start. The script clears all load inputs, waits 12 seconds for the model to settle, then writes one 30-minute timestep every 2 seconds.

Time Mapping in HIL Simulation

In this module, the five-day load and PV dataset contains **240 half-hourly intervals** (48 intervals/day $\times$ 5 days). Rather than running the simulation for five real days, each 30-minute interval is **mapped** to **2 seconds** of HIL simulation time.

What the console output tells you

Each line shows the **Step** (out of 240 total), the **Date and time** interval, the total **Load** (kW), total **PV** (kW), and **Net Load** ($P_{net} = P_{load} - P_{PV}$). A negative P_{net} at a node indicates reverse power flow, e.g., power flows from the customers at the node to another node along the Monash IEEE 13 Node Feeder. In contrast, a positive P_{net} indicates power flow from another node on the Monash IEEE 13 Node Feeder to the customers connected to the node with a positive P_{net} .

Note: Computing Interval Overrun (COI) Warning

Once the simulation starts, you may see a **COI (Computing Interval Overrun)** flag turn red at the bottom of the HIL SCADA window, along with a warning similar to Figure 21. This indicates that the signal processing computation time briefly exceeded its allocated time slot — it does **not** mean your playback script or setpoints are wrong, and it will not affect the validity of your five-day playback results.

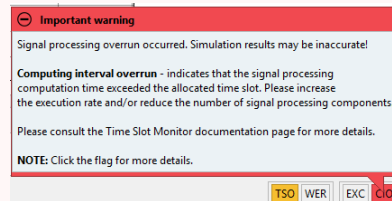


Figure 21: COI overrun warning shown in HIL SCADA.

Clicking on the **COI** flag opens the **Time Slot Monitor**, shown in Figure 22, which displays the CPU utilisation for each execution rate on the HIL device.

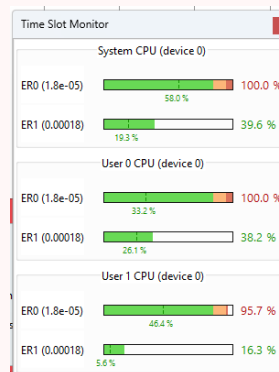


Figure 22: Time Slot Monitor showing System/User CPU utilisation.

To clear the warning, **right-click** on the **COI** flag and select **Reset Flag**  **Reset flag**. The warning indicator will disappear.

5. **Observe voltages and power flows across the Monash IEEE 13 Node Feeder in HIL SCADA. This is worth 50% of your assessment grade.**

As the playback runs, the **Measured Voltage Across Nodes** plot and the **Power** flow plot on the main SCADA panel will start to populate with live data, as shown in Figures 23 and 24.

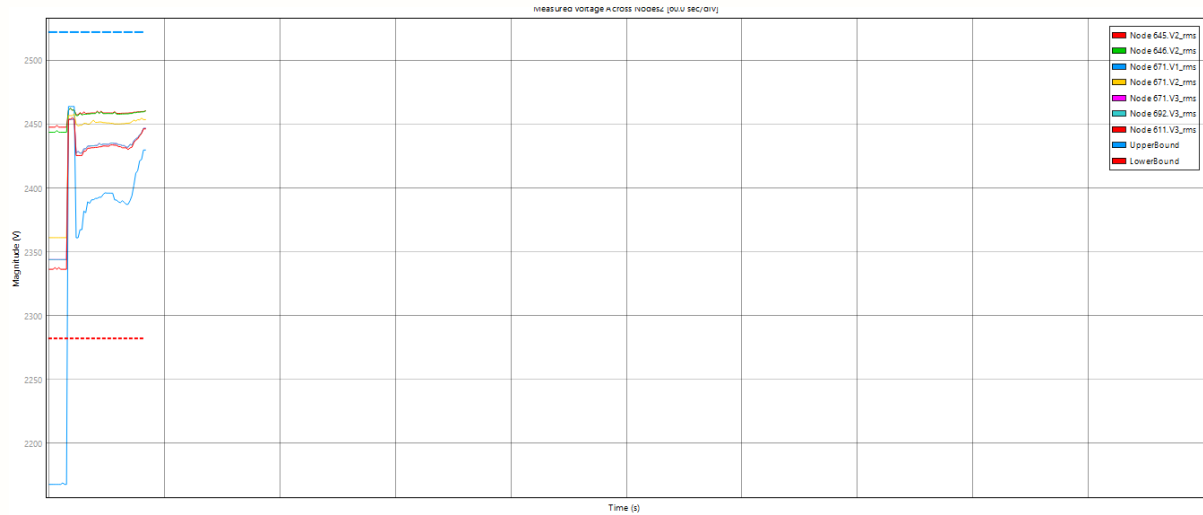


Figure 23: Measured node RMS voltages (V) during the five-day playback. The dashed blue line (**UpperBound** = 2522 V) and dotted red line (**LowerBound** = 2282 V) correspond to $\pm 5\%$ of the 2401 V nominal line-to-ground voltage. Note that this is an incomplete simulation, as the complete simulation will form 25% of your assessment.

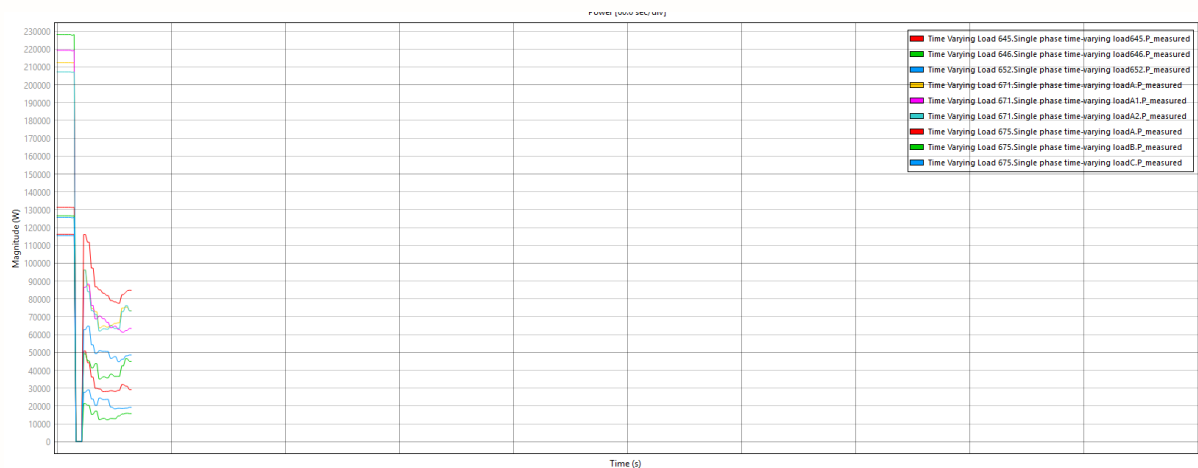


Figure 24: Measured active power (in W) at each time-varying single-phase node during the five-day playback. This is an incomplete simulation, as the complete simulation will form 25% of your assessment.

6. Wait for playback to complete.

The full 240-step playback takes approximately **9 minutes** at 2 seconds per step. The console will print **Playback complete.** when finished. Do not stop the simulation before playback completes. The console as shown in Figure 20 will close.

7. Stop the simulation.

Click the  **Stop** button to halt the simulation.

How to Stop the Playback Script

The playback script runs in a separate Windows console window. The HIL SCADA **Stop** button stops the simulation, but it does **not** terminate the playback script. Use the following steps to terminate the script:

- Go to the console window that is printing the playback output.
- Click inside the console window and press **Ctrl + C**.
- A message will appear: **Terminate batch job (Y/N)?**, as shown in Figure 25.
- Press **Y** and then press **Enter**.

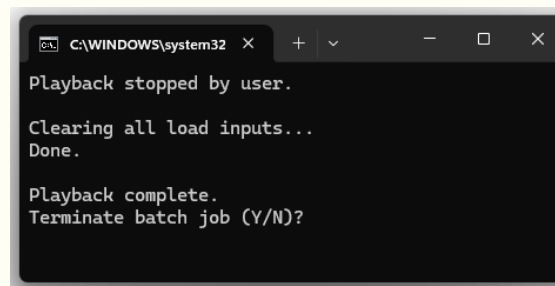


Figure 25: Console prompt displayed after pressing **Ctrl + C**.

Before restarting another simulation, make sure there is no playback console still running from the previous simulation. If you start the simulation again while an old console is still running, the new run may produce an error.

8. Locate the output CSV files.

Navigate to your **Project Directory** in File Explorer. You will find two new CSV files:

File	Contents
measured_voltages.csv	Per-step RMS voltage readings (V) from all monitored nodes. Columns: step , date_label , profile_time , then one column per node-phase voltage signal.
measured_powers.csv	Per-step measured active power (W) from all controllable load components. Columns: step , date_label , profile_time , then one column per node-phase load signal.

Note: Files Are Never Overwritten

Subsequent runs create new timestamped files automatically (e.g., `measured_voltages_20260622_143012.csv`). Your original files are always preserved.

9. Plot the results.

Open the two CSV files in Python, Excel, or MATLAB and produce:

- Measured power vs. time step:** all single-phase nodes will have active power traces on a single set of axes, with a legend.

- b. **Measured voltage vs. time step:** all node RMS voltage traces on a single set of axes, with horizontal dashed lines at 2522 V (corresponding to the upper voltage limit) and 2282 V (corresponding to the lower voltage limit).

7 Grade Sheet

Student Name:

Student ID:

Final Grade: /100

Real-time Simulation Results (Grade: /50)

Presented three real-time simulation results for the 5-day playback. No part-marks awarded — they either get the full grade or 0 for each item below.

1. Real-time RMS line-to-ground voltages for each single-phase node operating at the nominal 2401 V.
Grade: /20
2. Real-time RMS line-to-ground voltages for each single-phase node operating at the nominal 277 V.
Grade: /5
3. Real-time active power at all single-phase nodes with time-varying load.
Grade: /25

Oral-based questions (Grade: /50)

Answers to the questions below should be verified using the real-time simulation plots, corresponding to the 5-day playback. The answers need to be completed during the 10-minute allotted time. No part-marks awarded — they either get the full grade or 0 for each item below. Full marks demonstrate sufficient understanding. That is, if you think the student only partly answered the question and feel that a 50% grade is warranted, then this would be sufficient to award a full mark for the relevant question. Otherwise, no grade should be awarded.

1. Which nodes violate the upper or lower voltage bounds?
Grade: /10
2. When do the voltage violations occur?
Grade: /5
3. What is the worst-case voltage violation for each node where a voltage violation occurs?
Grade: /5
4. Which nodes exhibit reverse active power flow?
Grade: /10
5. When does the reverse power flow occur?
Grade: /5
6. What is the largest reverse power flow for each node where reverse power flow occurs?
Grade: /5
7. When is the total feeder net load highest across the five-day playback?
Grade: /4
8. Which nodes contribute most to the peak load for the feeder?
Grade: /4
9. How does the peak load period influence voltages across the feeder?
Grade: /2

References

- [1] Typhoon HIL, Inc., “Real-time simulation and testing.” <https://www.typhoon-hil.com/community/our-difference/real-time-simulation-and-testing/>, 2025. Accessed: April 2026.
- [2] Typhoon HIL, Inc., “HIL101 real-time simulator.” <https://www.typhoon-hil.com/products/hil-simulator/hil101/>, 2024. Accessed: April 2026.
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